

Branch and Bound

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Branch and Bound is a systematic algorithmic framework for solving optimization problems by intelligently exploring and pruning the solution space to find the optimal solution.

Introduction

Branch and bound algorithms are used to find the optimal solution for combinatorial, discrete, and general mathematical optimization problems.

- **Branching:** The main problem is broken into smaller subproblems, forming a tree-like structure where each node represents a partial solution.
- **Bounding:** For each node, the algorithm computes an upper or lower bound on the best possible solution within that subproblem. If this bound is worse than the current best-known solution, the node is pruned (i.e., not explored further).
- **Pruning:** By eliminating branches that cannot yield better solutions than the current best, the algorithm reduces the number of computations, enhancing efficiency.

Example Applications

- 0/1 Knapsack Problem
- Traveling Salesman Problem (TSP)
- Job Assignment Problem
- N-Queens Problem
- 15-Puzzle Problem

15-puzzle problem

- ▶ The 15-puzzle is a classic **sliding tile puzzle** consisting of a 4×4 grid with 15 numbered tiles and one empty space.
- ▶ The objective is to rearrange the tiles from a given initial configuration into a goal state (usually ascending order) by sliding tiles into the empty space.

Problem Characteristics

- ▶ State Space: Each configuration of tiles is a state. There are $\{16!\}/\{2\}$ valid states due to parity constraints.
- ▶ Moves: Up, down, left, right — only adjacent tiles to the blank space can move.
- ▶ Goal State: Typically tiles numbered 1 to 15 in order, with the blank in the bottom-right corner.

Traveling Salesman Problem using Branch And Bound

- ▶ Given a set of cities and distance between every pair of cities, the problem is to find the shortest possible tour that visits every city exactly once and returns to the starting point.

Traveling Salesman Problem using Branch And Bound

- ▶ The Travelling Salesperson Problem (TSP) is a classic optimization challenge where the goal is to find the shortest possible route that visits each city exactly once and returns to the origin city.
- ▶ The Branch and Bound method is a powerful technique to solve TSP more efficiently than brute force, especially for small to medium-sized instances.
- ▶ Key Concepts
 - Branching: Divide the problem into smaller subproblems (partial paths).
 - Bounding: Calculate a lower bound on the cost of completing each partial path.
 - Pruning: Eliminate paths that cannot yield a better solution than the current best.

Steps in the Algorithm

- ▶ Start with a cost matrix representing distances between cities.
- ▶ Reduce the matrix to calculate an initial lower bound (subtract row and column minimums).
- ▶ Branch by choosing a city to visit next and update the cost matrix accordingly.
- ▶ Bound: For each branch, compute the new lower bound.
- ▶ Prune: Discard branches with bounds higher than the current best solution.
- ▶ Repeat until all cities are visited and the optimal path is found.